

Exploring the Efficacy of a Chatbot Training Application in Alleviating Graduate Students' Public-Speaking Anxiety During Q&A

Hannah La

School of Information
Rochester Institute of Technology
Rochester, New York, USA
hl2291@rit.edu

Ha-Kyung Kong

School of Information
Rochester Institute of Technology
Rochester, New York, USA
hidy.kong@rit.edu

Ziming Li

School of Information
Rochester Institute of Technology
Rochester, New York, USA
zl1398@rit.edu

Roshan L Peiris

School of Information
Rochester Institute of Technology
Rochester, New York, USA
roshan.peiris@rit.edu

Abstract

Public-speaking anxiety (PSA) is a significant challenge for university students, affecting both their academic and professional success. Our work explores the potential of a large-language-model (LLM)-driven chatbot training tool to mitigate PSA, addressing the common negative perception of public-speaking skills among students. While most existing studies focus on anxiety during scripted presentations, this formative diary study examines the effectiveness of two versions of a chatbot application (virtual reality (VR) and desktop) in reducing anxiety during question-and-answer (Q&A) sessions. Our study aims to assess the impact of the chatbot training tool on graduate students' PSA, explore its practical applications, and compare the effectiveness of both versions against each other. The findings offer valuable insights for developing more effective and sustainable public-speaking training interventions.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; **User studies**.

Keywords

Virtual reality, Public-speaking anxiety, Human-AI interaction, Generative AI, Large language models

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1 Introduction

Public presentations are crucial for university students, not only to showcase their academic work but also for their professional success, such as during job searches. However, many students report experiencing public-speaking anxiety (PSA) while presenting [3, 11]. A 2017 study [8] found that 63.9% of college students experience fear of public-speaking, while 89.3% expressed a strong desire for public-speaking improvement classes. Students often feel underprepared for presentations due to factors like the lack of oral presentation courses, real-time feedback from audiences, and suitable rehearsal environments [1, 15]. Researchers have also identified a correlation between performance quality and the time spent rehearsing aloud in front of an audience [14].

In addition to in-person practices, recent research explores using technologies such as virtual reality (VR) and artificial intelligent (AI)-driven agents that offer independent presentation and Q&A therapy in areas, including job interview training [4, 19, 24–26] and public-speaking [15, 17, 18]. Specifically, in the context of VR, Reeves et al. used 360-degree video in VR exposure therapy to reduce PSA, finding that this approach significantly decreased PSA, social anxiety, and fear of negative evaluation [22]. VR technologies can simulate immersive environments, such as a classroom or theater stage, to replicate public-speaking scenarios [2, 6, 7, 15, 23, 27, 28]. Meanwhile, AI-driven agents can act as audiences or instructors, providing feedback tailored to the presenter's performance [15, 21]. Recent studies have begun adopting large language models (LLMs) to power AI-driven agents, enabling them to adapt their responses and feedback more flexibly across a wide range of communication topics [10, 12, 13, 18]. This approach allows the agents to generate feedback tailored to different professional levels, simulating a more realistic feedback experience. For example, Li et al. developed a VR training platform integrating LLM-driven avatars to train autistic individuals in job-related communication skills across diverse scenarios [12, 13]. Park et al. proposed AudiLens which applied a platform for practicing public-speaking using LLM-driven agents as audiences [17]. Min et al. conducted a short-term evaluation to preliminarily test the feasibility of using LLM-driven virtual audiences in a VR-based platform

for practicing Q&A in public-speaking and creating a lifelike public-speaking environment [15]. While the study tested the feasibility of LLM-driven agents in assisting with public-speaking Q&A training, it lacked a longer-term evaluation on how such a platform could help reduce presenters' PSA levels, particularly during the Q&A process.

With these challenges in mind, we conducted a five-day explorative study that evaluates the potential effectiveness of such a platform in improving the anxiety a graduate presenter may experience during Q&A. Moreover, we compared the user experiences of being trained via VR versus desktop format, as both VR and desktop applications have been validated as sufficient training tools in the past [2, 4, 6, 9, 20, 22, 25], aiming to address **RQ1**: What kind of impact will a Q&A chatbot training application have on a graduate student's public-speaking anxiety and sense of preparedness? and, **RQ2**: As a public-speaking training tool, how does its VR format compare to the desktop version?

2 Implementation

2.1 Prototypes

Inspired by previous work [12, 15, 17], we developed a speech-based chatbot application compatible with Meta Quest VR headsets and Windows PCs, built using Unity Platform version 2022.3.7f1 and Oculus Integration Package version 54.1. Virtual characters leverage the Meta Avatars SDK, enabling realistic facial expressions, eye movements, and lip synchronization. The VR application transcribes the user's speech using OpenAI Speech-to-Text (Whisper). The transcribed text is sent to the GPT-3.5 Turbo model ("gpt-3.5-turbo-0125") via OpenAI API, and its response is converted into speech using Oculus Text-to-Speech (Voice SDK). The VR application features an immersive 360-degree classroom environment constructed from a real-world photo (See Figure 1(a)), while the desktop application simulates a Zoom meeting interface to increase immersion by recreating a Q&A scenario over a video-conferencing platform (See Figure 1(b)).

To provide the user with the option to practice speaking with different personalities [19], three audience difficulty levels that are made implemented: Friendly (Level 1), Down-to-business (Level 2), and Critical (Level 3). A modal displays the difficulty level and a brief description for each level the user tabs to, as well as global instructions for how to interact with the avatar and application. The user can freely switch between the three levels.

2.2 Prompt Difficulty Levels

The structure of all three difficulty level prompts, regardless of the audience member's demeanor, includes key elements to guide the Q&A session with the presenter: (1) defining the audience persona (from friendly to critical), (2) encouraging thought-provoking and presentation-relevant questions, (3) outlining interaction guidelines (e.g., asking one question at a time, avoiding self-answers), (4) allowing for follow-ups and constructive critique, steering conversations back to the presentation topic if needed, and (5) maintaining an active and engaging dialogue throughout.

Inspired by previous work [12, 29], prompts for each of the three difficulty levels followed a similar structure and outline roles for an audience member to engage with the presenter during a

Q&A session. Level 1 outlined a role for a friendly and forgiving audience member, aiming to encourage a free-flowing and low-pressure interaction that helped the presenter feel comfortable without the fear of criticism. Level 2 described an audience member with a more focused and professional demeanor, ensuring a more structured and goal-oriented interaction. Level 3 defined a role for a critical audience member, aiming to test the presenter's ability to perform under stress and criticism from audiences.

3 Methodology

3.1 Participants

Participants were recruited through multiple channels (text, emails, flyers, and word-of-mouth). A screening survey was sent to potential participants to gather demographic information, assess their anxiety levels during post-presentation Q&A, determine if they were currently working on a project presentation, inquire about their VR/desktop equipment, and check for any risk of VR sickness. Participants were selected if they: experienced moderate to very high anxiety, were working on or had recently completed a presentation (within the last 2 weeks), had regular access to a Windows computer or Meta Quest headset, and were not at risk of VR sickness. Six graduate students (3 male, 3 female), with anxiety levels ranging from moderate to very high, were recruited. Four students had a background in human-computer interaction, one had a background in instructional design, and one had a background in art therapy.

3.2 Experiment Setup

Prior to the test phase, each participant was required to submit a presentation script of around 350 words long so that the scripts could be inputted into the chatbot system; most participant submissions centered around their own graduate capstone research. During their initial sessions, conducted with the primary researcher present, participants completed a 5–10-minute pre-test questionnaire. This questionnaire assessed their public-speaking experience, reasons for experiencing PSA during Q&A, anticipated anxiety levels for upcoming in-class Q&A sessions, and preferred preparation methods. Participants then engaged in 15-minute tutorial sessions with their assigned VR or non-VR chatbot applications to familiarize themselves with the system before starting the practice period.

3.3 Experiment Design

The experiment employed a between-subjects, five-day diary study design in which three participants utilized the VR chatbot application to practice Q&A (P1-P3) and another three used the desktop version of the application (P4-P6). The study's design was influenced by several logistical factors, including a limited budget for participant compensation, time constraints, and the desire to enhance participant retention and adherence to the multi-day study protocol.

The first session for each participant was conducted in a lab on campus with the primary researcher present. For the following four sessions, participants used their assigned applications daily for 15 minutes at times and locations of their choosing. After using the applications daily, participants documented their feelings and

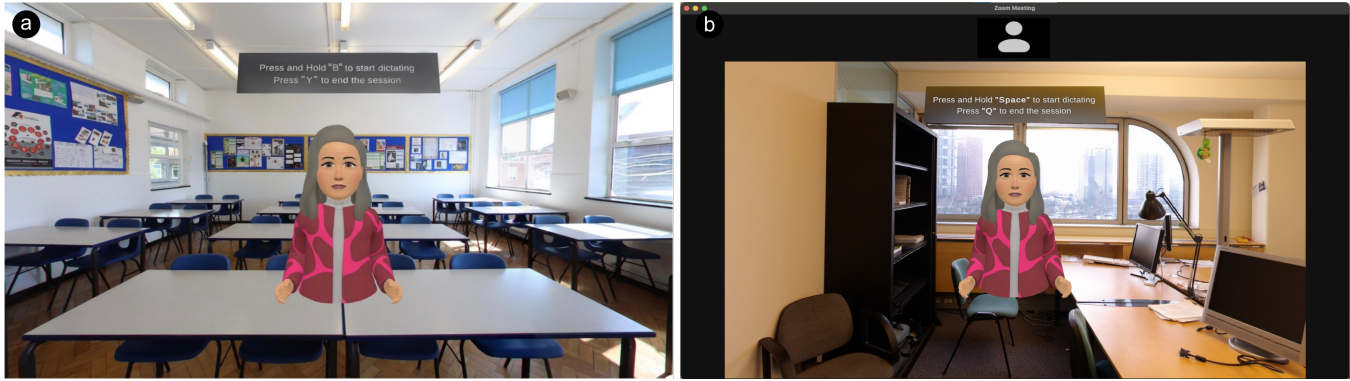


Figure 1: (a) Screenshot of the 360° virtual classroom in the VR chatbot application. (b) Screenshot of the mock Zoom interface in the desktop chatbot application.

experiences in journals, responding to open-ended questions and Likert-scale survey items designed to assess their experiences with the applications, anxiety levels, and preparedness throughout the study. Written transcripts of all system interactions were recorded. Upon completion of the study period, participants were interviewed for 30 minutes to discuss their subjective experiences with their assigned systems.

3.4 Data Analysis

Session transcripts and participant journals provided daily data, allowing us to track feelings and performance throughout the study. Written responses to open-ended journal questions were combined with interview data and analyzed using Braun and Clarke's thematic analysis method [5]. Interviews were transcribed, and both transcripts and journal entries were coded together. Initial codes were refined and collated through multiple reviews, leading to broad themes that captured user experiences, chatbot reception, and speaking performance outcomes. Quantitative data, such as Likert-scale responses, were also analyzed to identify trends over time.

4 Findings

4.1 Quantitative Data Analysis

Some of the objectives of this study were to explore how the chatbot training application impacted public-speaking anxiety and preparedness for Q&A, as well as to compare the effectiveness of the VR and desktop applications in reducing anxiety. To address these objectives, several journal survey questions were designed to measure these metrics.

Participants rated the chatbot's usefulness in identifying weaknesses in their presentations at 3.8 (SD = 0.2) and its overall utility in improving their presentations at 3.97 (SD = 0.2) on a 5-point Likert scale. Further analysis with a t-test did not reveal statistically significant changes during the five-day period.

Participants also rated their anxiety levels after each session using a 5-point Likert-scale, with the results for VR and desktop participants shown in Figure 2. For VR participants, the mean anxiety

ranged from 3.67 to 2.00; there is also a relatively strong linear relationship that suggests a downward trend over the five-day period. In contrast, the anxiety levels for desktop participants fluctuated between 3.33 and 2.66, with no significant changes observed.

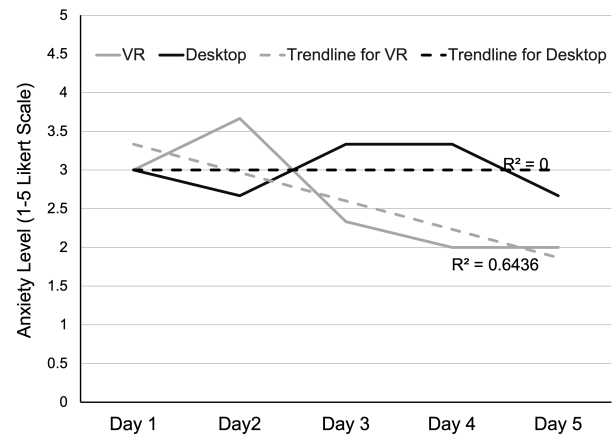


Figure 2: The average scores (on a 5-point Likert scale) showing VR and desktop participants' anxiety levels across five sessions.

4.2 Enhancing Preparedness Through Adaptive and Progressive Questioning

Combined journal and interview findings suggest that the chatbot offers advantages over traditional preparation methods. Unlike conventional approaches, it provides structured interactions which enhance the preparation process.

4.2.1 Initial Anxiety and Anxiety Reduction Through Repetitive Practice. Both VR and desktop participants initially experienced discomfort with the application, primarily due to unfamiliarity with the technology. However, as the study progressed, participants reported a decrease in their initial apprehensions and a gradual adaptation to the interfaces. P1, for example, described feeling "nervous" during

her first session but mentioned in her interview that she acclimated over time. Similarly, P4 initially felt “scared” when the chatbot asked him questions but eventually grew more comfortable, finding the chatbot’s questions helpful rather than intimidating.

4.2.2 Diverse and In-Depth Questions Leading to Insights on Improving Presentation Content. Participants were initially surprised by the depth of the questions the chatbot posed. The spontaneity of the questions helped participants like P2 feel like it “prepared me to answer on the spot and think and talk about it.” Moreover, the detailed nature of the questions encouraged participants to be more reflective and consider alternate perspectives that they had initially overlooked in their study designs. P1 gave an example of an oversight that the chatbot had unearthed, admitting, “[...] the cultural differences that people have when using gesture recognition was something that I didn’t really think about, which was that people do use technology differently in all different ways.” Though most participants did not update their scripts during the study, all reported that the chatbot helped them identify weak points in their presentation content.

4.2.3 Engaging With Higher Levels to Test Abilities. Participants approached the difficulty levels differently. Most participants showed curiosity about the progression and using the levels to challenge and extend their abilities. Of the participants that tried different difficulty levels, most began at Level 1 and gradually increased the difficulty over several days. As P2 noted, “I wanted to try different levels to know how the questions and the feedback from the chatbot would be. So even if there wasn’t enough time, I would select other difficulty levels in the later days.” Similarly, P6 noted, “For me, it was more like a challenge to answer all the questions. [...] I wanted to finish all three levels no matter what.”

However, some participants preferred specific levels or chose to remain on one throughout the study, often because they felt certain levels better matched their real-life Q&A expectations. Here, P5 indicated, “In Level 1, she [chatbot] was asking really easy questions, and I didn’t feel very productive. When I got to Level 2, I felt like that was the sweet spot because the questions were meaningful and not too long like Level 3.” Meanwhile, P3 commented, “I stayed with Level 1 the whole time because I estimate the crowd I’ll be interacting with in the actual Q&A will be similar in difficulty.”

4.2.4 Allowing Time for Thoughtful Responses. A particularly valued feature mentioned by participants was the ability to pause before responding to questions, which they noted as significantly beneficial for improving their answers and reducing anxiety. P1 noted the flexibility this feature offered: “It didn’t force me to answer right away. So I had the ability to click on the button and then respond to it like when I was ready to answer.” P4 provided further insight into how this feature helped improve their communication: “It gave me the opportunity to answer better and stop myself and then speak at least more intelligently as compared to before.”

4.3 Immersive Aspects of the Chatbot Need Tuning

4.3.1 Comparing the Immersiveness of the VR and Desktop Applications. Participants generally reacted positively to the VR environments and acknowledged the enhanced immersion that these

settings can offer. However, P4 suggested improvements that could make these simulations more realistic and beneficial: “[...] you could add a bunch of people sitting in the audience to give people familiarity with situations where it’s kind of crowded. So that definitely will help reduce anxiety with more exposure to people and those kinds of crowded environments.”

Despite enjoying enhanced immersion, some participants questioned the necessity of a VR setup for effective training. P1 expressed a preference for simplicity and accessibility, stating, “I would have been fine with either doing it on my laptop and just kind of having it pop up on the screen like ChatGPT and answering the questions.” P3 also pointed out the over-complication of VR setups during her interview: “I don’t necessarily think I need the VR setup to feel like I’m getting a lot out of it. So if it was more accessible in that way where I didn’t have to set up my VR and go into the app and do the whole thing. If it was just something where I could just open it and start talking to it, I feel like I would use it more often for smaller things, not just, like, a big presentation.”

These perspectives suggest that while VR can offer unique benefits, the technology may not be essential for all types of training, especially when simpler, more accessible alternatives could suffice.

On the other hand, desktop participants did not remark on the mock Zoom interface, suggesting it had little impact on their sense of immersion. This absence of feedback may indicate that the desktop application was less engaging compared to the VR application, which received unsolicited positive comments from participants.

4.3.2 Unrealistic Questions or Lines of Questioning. While participants were generally satisfied with the chatbot’s performance, they still drew attention to areas they felt warranted improvement. Some found the chatbot’s questioning techniques problematic, describing the questions as “huge” (P6) and “long-winded” (P5), which led to confusion and frustration. P5 specifically noted difficulties with the lengthy questions in Level 3, explaining that the captions were broken into two parts, making it difficult to remember the first half by the time the second half appeared.

Furthermore, some participants felt that the chatbot sometimes focused on certain topics to a degree of irrelevancy. Although they appreciated the opportunity to refine their answers when the chatbot revisited topics across sessions, the repetitive nature of some out-of-scope and follow-up questions caused irritation; this most often occurred when the chatbot asked about future plans or tangential work.

4.3.3 Unnatural Avatar Design and Implementation. The design and implementation of the virtual avatar significantly influenced participants’ immersion in the VR training experience. Feedback highlighted the avatar’s physical appearance, gestures, and voice quality as key factors that either enhanced or detracted from the simulation’s realism and user engagement.

However, not all feedback was critical. P3 noted that even an imperfect avatar could still maintain a sense of presence and interaction: “Even though the person doesn’t necessarily look realistic, it still felt like there was a person there.” This perspective highlights the potential for avatars to support effective training despite design flaws by preserving a sense of human interaction in the VR environment.

Additionally, participants emphasized the need for more expressive avatars. P2 observed that limited emotional expression and human-like gestures reduced the authenticity of the interactions, suggesting improvements in these areas could enhance engagement. Similarly, P3 commented on the avatar's facial expressions: "It would be interesting to have its facial expressions do more like actual responding. A couple of times, I was distracted by that, too."

The avatar's voice quality was another critical factor affecting immersion. Here, P2 and P5 described the voice as monotonous and lacking natural dynamics, suggesting the integration of a more sophisticated text-to-speech system to create a more lifelike conversation experience.

4.4 Tough Audiences May Hinder More Than Help

4.4.1 Anxiety and Challenge in Advancing Through Levels. Some participants expressed mixed feelings about progressing through increasingly difficult levels of the training scenarios. P3 experienced anxiety at the prospect of engaging with tougher levels, stating, "[...] it actually was giving me a little anxiety at the thought of trying the tougher levels, so I didn't." This anxiety was significant enough to deter her from exploring these levels purely for exploration's sake.

P6 was unique in that she provided a more detailed account of her experience with the training scenarios, indicating that completing the lower levels brought a sense of accomplishment, particularly since feedback was positive: "It felt good to reach the end of all the questions in Level 1 and Level 2 [...] and (if) at the end I received feedback that I was in a good place or something like that—that was kind of nice to hear."

However, P6 also highlighted a stark contrast in Level 3, where the avatar's unfriendly tone made them reluctant to engage: "Level 3—her tone... I guess it's just not friendly at all, so I didn't really feel like answering."

4.4.2 The Need for Reassurance. The tone of the avatar significantly impacted how some participants, especially P6, felt about the training on different days, especially for those experiencing fluctuating anxiety levels. P6 articulated a need for ongoing reassurance within the training scenarios to combat feelings of inadequacy or anxiety: "I think one thing that I feel when it comes to having anxiety in these kinds of situations is that reassurance is always a good thing so that you know that, 'Okay, you're not doing too bad at it.'"

Although this feedback was particular to P6, it nevertheless sheds some light on the importance of designing training tools that not only challenge users but also provide emotional support during the learning process. Additionally, P6 suggested incorporating chatbot feedback in the form of tips, advice, or sample answers to help users learn from their mistakes and better understand how they could have "handled the question better" or "phrased something differently."

5 Discussion

5.1 Viability of Our Training Application as a Practical Solution

One objective of this study was to examine the impact of prolonged use of the chatbot training application on graduate students' public-speaking anxiety (RQ1). Findings indicated that participants appreciated several features, including the diversity of questions, revisitation of key topics across sessions, the ability to adjust difficulty levels, and the flexibility to answer at their own pace. However, participants identified barriers to progress, including overly complex or irrelevant chatbot questions and the hyper-critical personality of the Level 3 avatar, which discouraged serious attempts. Two key insights emerged from these observations: (1) improved prompt engineering is needed to ensure questions are focused, relevant, and avoid redundant topics; and (2) while challenging interactions are valuable for replicating Q&A experiences, integrating constructive feedback and real-time performance reports could better support users' development [7, 16].

Interview responses revealed that participants viewed the chatbot as a valuable tool for ongoing practice, appreciating its capacity for repetitive use and immediate verbal feedback. Many expressed interest in integrating it into their preparation routines for future presentations and highlighted additional applications, such as improving interview skills. Journal and transcript data supported these perceptions, showing satisfaction with the chatbot's effectiveness in refining presentation content and indicating gradual improvement in participants' clarity and confidence over time. These findings emphasize the chatbot's potential for long-term use and broader practical applications beyond the study.

5.2 The Potential Superiority of Training in VR

This study also aimed to compare the effectiveness of the VR and desktop applications (RQ2). Results indicated a declining trend in anxiety among VR users, suggesting that VR's immersive nature may help desensitize participants to public-speaking scenarios. The sense of "presence" created by the 3D audience and setting likely provides a more engaging and grounding experience. These findings align with previous research on the effectiveness of immersive environments for anxiety-inducing tasks such as public speaking [2, 6, 20, 22]. However, some VR participants questioned the necessity of these added benefits, citing insufficient immersion or the complexity of installation as drawbacks.

In contrast, desktop users benefited from the convenience and accessibility of the application but did not experience the same level of anxiety reduction or immersion. The lack of significant anxiety reduction in desktop settings suggests limitations in replicating the stress of live Q&A experiences. Interestingly, while desktop participants reported feeling less anxious during interviews, their journal data showed minimal changes in anxiety levels. This discrepancy may stem from: (1) a general reduction in task-related anxiety offset by the stress of attempting higher difficulty levels later in the study, or (2) external distractions during sessions (e.g., as reported by P6), which may have influenced their focus.

5.3 Limitations and Future Work

This study has several limitations that inform directions for future research. First, the results are not generalizable due to the small participant sample, which constrained the scope of analysis to basic descriptive statistics. Future studies with larger samples could help identify more usage patterns and insights.

Second, the reliance on self-reported anxiety levels in our user studies could introduce potential biases. Future studies incorporating physiological measures could offer a more objective assessment of anxiety and performance over time. Moreover, expanding this research to include real-world evaluations with live audiences or public-speaking experts could provide a more comprehensive measure of the intervention's effectiveness.

Despite these limitations, the study demonstrates the chatbot's potential as a practical training tool for improving preparedness and reducing anxiety in Q&A contexts. The comparative advantage of VR over desktop environments highlights the need for further investment in VR training technologies, particularly for training scenarios that require high engagement and realism. Future efforts should focus on enhancing the avatar's physical and behavioral realism with more nuanced real-time feedback mechanisms.

Beyond academic settings, the application shows promise for broader public-speaking contexts such as interviews, conference talks, and media training. Continued development and validation could advance understanding of anxiety management and promote technology-assisted learning across diverse domains.

6 Conclusion

To reduce graduate students' PSA and enhance their preparedness for Q&A sessions, our study evaluated the use of an LLM-driven chatbot application in both VR and desktop formats through a 5-day diary study with six participants. The results suggest that the application boosts confidence and readiness by providing dynamic, adaptive questioning that mirrors real-life interactions. Notably, the VR format appeared to be more effective at reducing anxiety than the desktop format, supporting the potential of immersive, interactive training tools to more effectively mitigate PSA.

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